

Ever since I was 17 and decided that I should do a Bachelor's degree in Physics, research has become a passion for me. In my undergraduate years, experimental physics was of much fascination to me, and towards the end of my degree, I started actively participating in research in the very same laboratory where, later on, I did my Master's. This early exposure to research not only set in stone my interest in the field but also set the stage for my future academic and professional journey into the field of material science.

In the Master's in Materials Science and Engineering, I had specialized in the characterization and application of 2D materials, which is more thrilling to me because of their promising potential in different state-of-the-art technologies. More precisely, I worked on the development and characterization of nanocomposites considering applications for devices from sensors, organic photovoltaics, and transistors to light-emitting diodes. One big research project I pursued was based on an aqueous solution nanocomposite resulting from two commercial materials: the conductive polymer composite PEDOT:PSS and 2D-MoS₂. In this work, the primary focus was to study the suitability of this nanocomposite for use as transparent electrodes in organic photovoltaic devices. Integration with such materials had provided a great avenue for the exploration of properties of 2D materials toward realistic applications.

Having been part of a research group dealing with nanostructured materials, I actively participated in the characterization of the composite. The characterizations of my work included thorough morphological, electrical, and optical characterization. The advanced techniques employed in this work include SEM and TEM for morphological analysis, electrical conductivity measurement, and temperature-dependent electrical conductivity to test the electronic properties of the material. Ultraviolet-Visible spectrophotometry and Raman spectroscopy were employed for optical characterization in the probed optical properties of the material and, in turn, confirm its suitability for photovoltaic application.

First, the relevance of the present work was pointed out when I had the opportunity to present my findings at two international conferences, B-MRS-XXI and B-MRS-XX, and in one Brazilian conference on carbon nanostructures. The exposure to the global research community not only helped me hone my scientific communication but also further strengthened my commitment to making a contribution to advances in material science. Apart from this, my current working is under development into a paper that is currently in the submission process for publication.

Post-master's, I extended my experience by joining SENAI Institute for Innovation-ISI, Brazil, to develop lithium-ion batteries. This provided an avenue for diversification into energy storage research. My core role in the project was to synthesize and characterize a lignin-based anode for lithium-ion batteries, the project had, in the general framework, the aim to connect materials science with sustainable solutions for energy. Further laboratory experience involved learning and applying several new characterization techniques: DLS, rheology, X-ray diffraction, and cyclic voltammetry.

At ISI, I also had the chance to receive valuable insight into industry related to research, therefore giving a broader perspective on challenges and opportunities regarding scaling up research from the laboratory to production. The experience whetted not only my practical skills but furthered my comprehension of advanced materials regarding real-world application and deepened my commitment to translation of fundamental research into tangible innovation.

Apart from my research experience, the opportunity to share knowledge and contribute to the growth of others has always meant much to me. During my undergraduate study, I conducted tutoring for two semesters for students in Pre-Calculus, a process that helped me in developing my teaching and communication skills. Later, when I was taking my Master's, I had a chance to share the same class of experimental physics with my academic adviser. There, in the lab, I could assist students in conducting experiments and explaining theoretical material. This provided a good opportunity for me to reinforce my knowledge while honing my skills in explaining what might seem complicated in more accessible ways.

In the long run, I am looking forward to continuing my research in 2D materials and devices at EPFL-a place which I strongly believe offers the best environment for advanced studies in Materials Science and Engineering. An opportunity to work with Professor Andras Kis, whose expertise greatly aligns with my

research interests, would be a great privilege. I have already approached him, and I am excited about the opportunity for working with him on state-of-the-art projects in nanomaterials and their applications in electronic devices.

In short, my academic background is comprehensive, including undergraduate research, a Master's in Materials Science, and industrial experience. It thus prepared me with diverse skills and a deep understanding of materials science. These experiences, combined with my curiosity and drive for the advancement of scientific knowledge, make me confident in succeeding in my PhD studies. I look forward to this new chapter in my career, which I hope to establish in the fast-growing area of 2D material research at EPFL.